CMPn332 – CMPN332 – CMP432

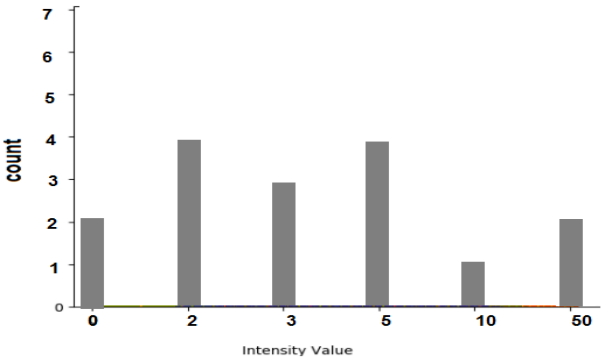
Histogram: is a graph, which gives you an overall idea about the intensity distribution of an image.

For the following image matrix find the histogram.

$$\begin{bmatrix} 0 & 50 & 2 & 3 \\ 5 & 5 & 2 & 2 \\ 5 & 2 & 3 & 5 \\ 10 & 50 & 3 & 0 \end{bmatrix}$$

Answer:

Gray_level	Count
0	2
2	4
3	3
5	4
10	1
50	2



Techniques used in image enhancement are:

- **Histogram Equalization**:
  It is a simple way to improve dark or washed-out images (low-dynamic range) by Redistributes pixel intensities across the entire range of available values based on CDF.
  S= CDF(r)×(L−1)
- **Contrast Stretching**:
  preferred for enhancing images with a known intensity range needed to be enhanced.

$$\alpha = \frac{s_1 - 0}{r_1 - 0}, \beta = \frac{s_2 - s_1}{r_2 - r_1}, \Upsilon = \frac{L - 1 - s_2}{L - 1 - r_2}$$

$$s = \begin{cases} \alpha * r & 0 \le \boldsymbol{r} < r_1 \\ \beta\,(r - r_1) + s_1 & r_1 \le \boldsymbol{r} < r_2 \\ \Upsilon(r - r_2) + s_2 & r_2 \le \boldsymbol{r} < L - 1 \end{cases}$$

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A 6-bit 50x80 image has the following intensities frequencies. Find out the equalized histogram gray levels for the image

Gray level	Count
29	1500
30	1000
15	100
22	500
42	900

Answer:

- A 6-bit →L=2^6 = 64
- 50x80 → M*N=50*80=4000
- equalized histogram gray levels for the image

Gray level r_k	Count n_k	Probability $P_r(r_k)=\frac{count}{M*N}$	cumulative probability= CDF=$\sum_{j=0}^{k} pr(rj)$	Equalized histogram= $(L-1)*\sum_{j=0}^{k} pr(rj)$
r0=15	100	0.025	0.025	2
r1=22	500	0.125	+= 0.15	9
r2=29	1500	0.375	0.525	33
r3=30	1000	0.25	0.775	49
r4=42	900	0.225	1	63

The histogram equalization stretch out the intensity range from 15:42 to become 2:63 which improve the image contrast

A 3-bit 4x4 image has the following intensities; Find out the output gray levels after applying contrast stretching. Using r1=3, r2=5 s1=2 and s2=6

```
[4, 3, 5, 2;
3, 6, 4, 6;
2, 2, 6, 5;
7, 6, 4, 1]
```

Answer:

$$\alpha = \frac{s_1 - 0}{r_1 - 0} = 0.66,\ \beta = \frac{s_2 - s_1}{r_2 - r_1} = 2,\ \Upsilon = \frac{7 - s_2}{7 - r_2} = 0.5$$

$$s = \begin{cases} 0.66 * r & 0 \le r < 3 \\ 2 * (r - 3) + 2 & 3 \le r < 5 \\ 0.5 * (r - 5) + 6 & 5 \le r < 7 \end{cases}$$

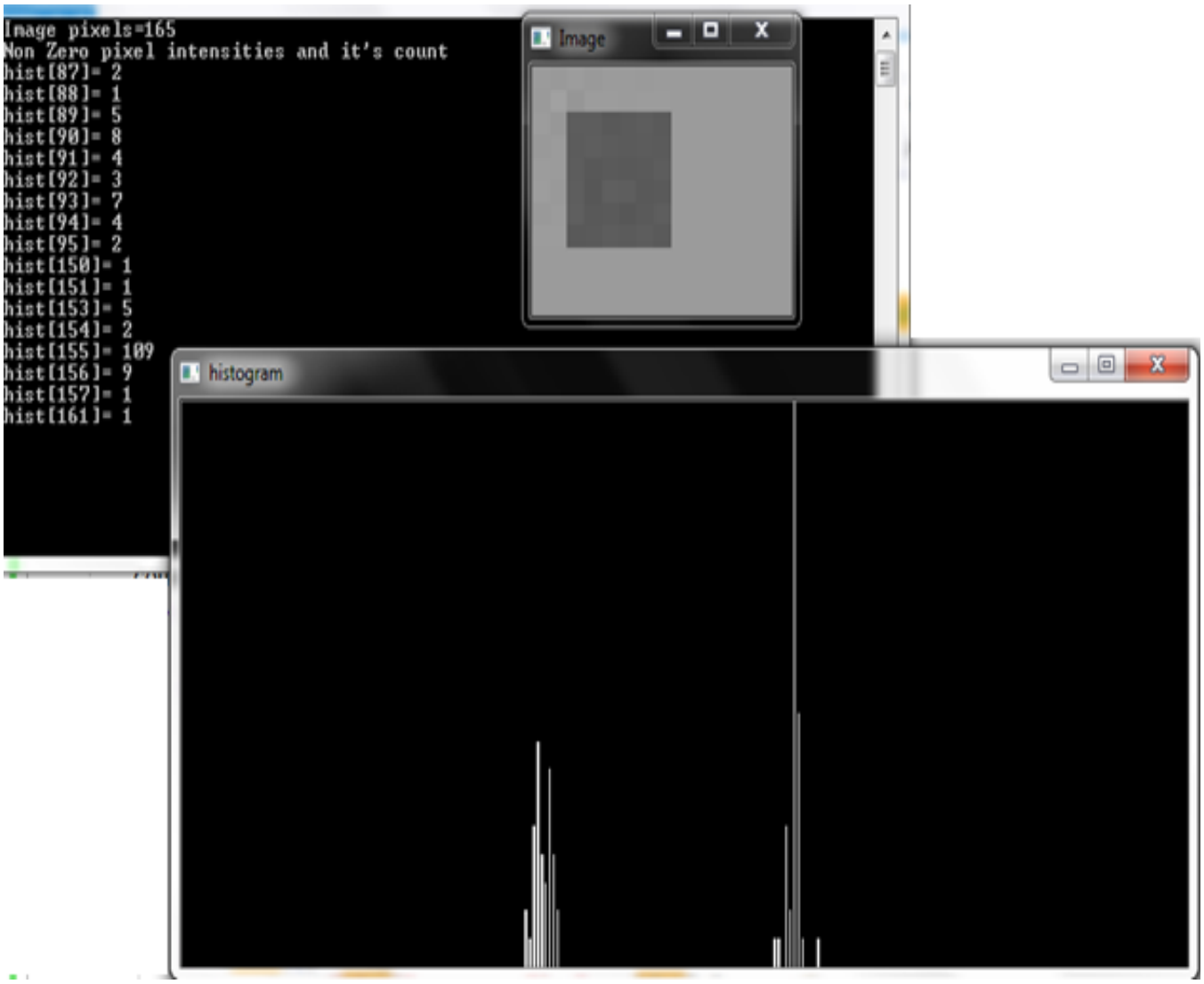
[4,	2,	6,	1;
2,	7,	4,	7;
1,	1,	7,	6;
7,	7,	4,	1]

Lab

Histogram

```
 Mat src3 = imread("histimage.jpg", 0);
Mat dst = src3;
namedWindow("Image", 0);
imshow("Image", src3);
waitKey(0);
cout << "Image pixels=" << src3.total() << endl;
float hist[256] = { 0 };
cout << "Non Zero pixel intensities and it's count" << endl;
   for (int g = 0; g < 256; g++)
       {
       for (int r = 0; r < src3.rows; r++)
         for (int c = 0; c < src3.cols; c++)
              {
              if (src3.at<uchar>(r, c) == g)
                 hist[g] = hist[g] + 1;
              }
              if (hist[g] != 0)
              cout << "hist[" << g << "]= " << hist[g] << endl;
       }
/// /////////////////object detection/////////////////
  for (int r = 0; r < src3.rows; r++)
       for (int c = 0; c < src3.cols; c++)
       {
       if (src3.at<uchar>(r, c) >= 150)
         dst.at<uchar>(r, c) = 255;
       }
       namedWindow("object", 0);
       imshow("object", dst);
```

```
        waitKey(0);
////////////////visualizing histogram//////////////////////////
Mat hist_img(20, 510, CV_8UC1);
hist_img = 0;
for (int s = 0; s < 256; s++)
        {
        float bin_val = hist[s];
        line(hist_img, Point(s * 2, 20), Point(s * 2, 20 - bin_val), cvScalar(255), 1);
        }
        namedWindow("histogram ", 0);
        imshow("histogram ", hist_img);
        waitKey(0);
        }
```



Histogram Equlization

```
#include<opencv2/opencv.hpp>
#include"iostream"
using namespace std;
using namespace cv;
int main()
   {
      Mat src3 = imread("hist.jpg", 0);
       Mat dst3;
       namedWindow("Src Image", 0);
       imshow("Src Image ", src3);

       equalizeHist(src3, dst3);

       namedWindow("histogram equlization", 0);
       imshow("histogram equlization", dst3);
       waitKey(0);

  }
```

